Kyoomin Kim

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Experience

Arista Networks June 2025 - Current
Software Engineer *Santa Clara, CA*

- Automating generation of platform code for Arista switches in Microsoft's SONiC OS by translating EOS switch data with Python scripts and C++ APIs, achieving 60% platform code coverage of 2500+ LOC
- Designing dynamic speed change support during runtime for switches in SONiC by pre-computing gearbox mappings and optic tuning values per speed mode from switch data
- Adapting EOS test framework for cross-platform use in testing link flaps, speed changes, etc. by abstracting away redundant logic and putting OS-specific code in wrappers, expanding test coverage by 20%
- Diagnosing link-up failures and performance issues in 100+ high performance 800G switches by analyzing and correcting lane optical values or mappings

Falkor July 2024 - Aug 2024
Software Engineering Intern *San Diego, CA*

- Created recommendation system using cosine similarity in Django and numpy with 6000+ keyword parameters
- Migrated 10000+ LOC MVT site into React Native frontend and DRF API, cutting resource costs by 50%
- Implemented secure real-time chat rooms with Channels websockets using sanitized messages and rate limiting

TrustThink Jan 2024 - June 2024
System and Security Engineering Intern *San Diego, CA*

- Developed a FIPS 140-3 cryptographic API in C++ and OpenSSL that supports encryption, authentication, etc
- Utilized UART and TLS protocols to provide a security interface in C between an STM32 MCU and software
- Ensured 100% test code coverage for all libraries with GoogleTest for units and FuzzTest for memory checks

UCSD CSES Aug 2023 - Dec 2025
Software Engineering Lead *La Jolla, CA*

- Led a team of 5 developers and designers for the CSES WebDev website using React, Python and PostgreSQL

UC San Diego Mar 2023 - Mar 2025
Instructional Assistant *La Jolla, CA*

- Tutored for Computer Security (CSE 127) for topics of memory corruption, web attacks, encryption and networks

Education

University of California San Diego Sep 2021 - June 2025
Bachelor of Science in Computer Science *La Jolla, CA*

Coursework: Computer Security, Networks, Embedded Systems, Operating Systems, Parallel Computing, Algorithms
Extracurriculars: CSE Society (Board Member, Software Lead), ACM, Video Game Design, Varsity Esports
GPA: 3.70

Projects

Pathtracer | *C++, Embree, FreeImage, GLM*

- Developed a renderer with Embree that supports analytical lighting, ray/path tracing and photon mapping
- Used BVH heirarchy, kd-trees and multithreading for 99% optimization in computing approximated radiance

Lifesaver | *C#, Unity, Meta XR SDK*

- Engineered an MR app in a team of 4 for navigating sites with hazardous AQI via wayfinding and passthrough
- Ensured frictionless experience with voice recognition, hand tracking and symbolic input using Meta XR

Technical Skills

Languages: Python, C++, C, C#, Java, JavaScript, Typescript, HTML/CSS, R, SQL, Haskell, MATLAB
Frameworks: Express, React, OpenSSL, OpenGL, GoogleTest, FuzzTest, Unity, Embree, Django, JavaFX, JUnit
Developer Tools: Git, Linux, AWS, Docker, CMake, LLVM, Valgrind, GDB, Binwalk, VirtualBox, Gradle, Postman